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Reproductive characteristics, semen quality, seminal oxidative status, steroid hormones, sperm production efficiency of rabbits fed herbal supplements

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ABSTRACT

Today, orthodox medicine has almost exceeded its limits in resolving subfertility problems in animals, thus making phytomedicine a primary tool in the treatment of infertility. In this work, three herbal supplements obtained from freshly air-dried *Moringa oleifera, Phyllanthus amarus* and *Viscum album* leaves were evaluated to ascertain their comparative effect on the reproductive potentials of bucks. Sixty bucks were allotted four diets made up of standard grower rabbit ration without supplement and with 5% Moringa, Mistletoe and Phyllanthus supplementation for 84 days. Semen samples were collected from all bucks using artificial vagina, for semen quality and seminal oxidative stress markers. The organ weights, testicular and epididymal spermatozoa reserves were assessed to determine sperm production potentials using standard procedures. The result obtained revealed that spermatozoa concentration, progressive motility, curvilinear velocity, average path velocity and the amplitude of lateral head was significantly ($p < 0.05$) higher in the group fed with mistletoe supplemented diet compared to that recorded in the control groups. The inclusion of herbal supplements linearly ($p < 0.05$) increased the seminal total antioxidant activity with a corresponding decrease in the seminal lipid peroxidation across the herbal supplemented treatments compared to the control. The gonadal and extra-gonadal sperm reserves of bucks fed on the herbal supplements were depleted compared to bucks on control. Bucks on mistletoe supplementation rivalled the superior daily sperm production and testicular sperm reserve in bucks without herbal supplements. In conclusion, *Viscum album* supplements in bucks' diets encouraged daily sperm production, testicular sperm reserves, testosterone, as well as spermatozoa kinetics. The three herbal supplements did enhance semen oxidative stability.

1. Introduction

Sexual health is an essential component of an individual's quality of life and well-being [1]. Infertility has become a recurring problem among male and female individuals, with more than 90% of male infertility cases linked to low sperm counts, poor semen quality or both [2]. Traditional herbs have been used to improve sexual performance or to treat infertility [3,4]. Some medicinal plants are extensively used as an aphrodisiac to relieve sexual dysfunction, or as fertility-enhancing agents [5], to provide a boost of nutritional value, which aims at improving sexual performance.

Infertility has been associated with high oxidative stress, thus indicating that any group of infertile animal (up to 80%) is more likely to have high seminal oxidative stress [6], as an increase in the production of reactive oxygen species (ROS) determines semen characteristics and sperm-oocyte fusion. Cellular damage in the semen could be the result of improper balance between reactive oxygen species generation and scavenging activities [7]. Thus, the protective effect of antioxidant supplements against ROS-induced

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oxidative stress may be essential to improve the spermatozoa quality of bucks in the tropics [8]. Determinations of biochemical constituents of seminal plasma are needed for semen evaluation, as physical characteristics of semen alone are not entirely satisfactory for semen appraisal in current practice [7]. Contribution of seminal composition to the large variability in semen quality traits evaluates seminal inclusions of interest to improve semen output [9].

The use of herbal medicinal plants is becoming an object of increasing interest due to its food safety, just as the use of antibiotics is becoming ever restricted because of its adverse effect on animals, residues in animal products and the development of antibiotic resistance [10]. In Nigeria, several plants have been identified to have medicinal and nutritional potency, and has been used to enhance fertility in male and female animals [11].

Feeds with additives rich in antioxidant could play a significant role in improving the reproductive health of rabbits [12]. Herbal plants such as *Moringa oleifera*, *Phyllanthus amarus* and *Viscum album* have proven to have high therapeutic value and tends to increase an animal's general performance [10,13,14]**.** In comparison to commonly used conventional forages, they contain low protein, moderate fibre, and high minerals, flavonoids and alkaloids with the presence of several compounds in one plant produces a synergistic beneficial effect on the improvement of animal performance [15]. Some herbs have been reported to improve libido, sexual behaviour, mating performance and spermatogenesis. In contrast, others have been reported to balance the levels of hormone such as testosterone, luteinising hormone, and follicle-stimulating hormone in the reproductive axis of male and female animals [11].

The mechanism of action of some plants known to possess antifertility effect through their action on the hypothalamic-pituitary-gonadal axis or direct hormonal effects on reproductive organs is not precise [16]. Given the consumption of *Moringa oleifera, Phyllanthus amarus, and Viscum album* leaves for the maintenance of health and well-being, the comparison of the effect of the herbs on sperm production efficiency, semen quality and peripheral hormonal levels for the enhancement of male fertility is imperative.

Hence, the need for the comparative assessment of the herbs as feed supplement on reproductive organ weight, semen quality, seminal oxidative status, steroid hormones and sperm production potentials of rabbit bucks.

2. Material and methods

2.1. Experimental animals and management

This study was carried out with the approval of the institutional ethics committee for the care and the use of animal for research purposes in line with NIH guideline. *Moringa oleifera, Phyllanthus amarus* and *Viscum album* leaves used for this experiment was harvested within the Teaching and Research Farm of the Federal Polytechnic Ado-Ekiti, Ekiti State Nigeria. Fresh leaves were harvested and shade-dried until crispy to touch and ground, and after that, they were referred to as Moringa leaf meal (MoLM), Phyllanthus leaf meal (PLM) and Mistletoe leaf meal (MiLM) respectively.

Sixty mixed bred (Chinchilla x California x New Zealand White) rabbits aged six weeks and weighing 775 ± 54.23g were used for the experiment. The bucks were randomly assigned to four [4] dietary groups consisting of fifteen animals each, in a completely randomised design. Experimental diets were formulated and pelleted to meet the nutrient requirement of growing rabbits, as shown in Table 1 and were fed to bucks at 4% body weight. The experimental diets are as shown in Table 1; T1 basal diet without leaf meal, T2 basal diet with 5% Moringa, T3 basal diet with 5% Mistletoe and T4 basal diet with 5% Phyllanthus.

Table 1
Gross composition of experimental diets (g/100g).

Ingredient	**Control**	**Moringa**	**Mistletoe**	**Phyllanthus**
Rice bran	6	1	1	1
Salt	0.25	0.25	0.25	0.25
Wheat offal	5	5	5	5
BDG	5.0	0	0	0
Grower premix	0.25	0.25	0.25	0.25
Maise	25	25	25	25
Methionine	0.40	0.40	0.40	0.40
SBM	17.0	17.0	17.0	17.0
Lysine	0.10	0.10	0.10	0.10
Bone meal	1.0	1.0	1.0	1.0
Groundnut Hauluns	40.0	40.0	40.0	40.0
Moringa	0	10	10	0
Mistletoe	0	0	10	0
Phyllanthus	0	0	0	10
Vegetable oil	0.20	0.20	0.20	0.20
Total	**100**	**100**	**100**	**100**
Calculated nutrient composition				
Dry matter %	88.05	85.41	81.2	87.45
Crude Protein%	16.47	16.58	17.94	17.667
Digestable Energy kcal/kg	2721.6	2515.35	2479.6	2509.4
Ether Extract %	3.344	3.022	3.754	2.419
Crude Fibre %	16.5	17.325	15.378	16.32
Lysine %	1.0085	0.9385	0.9385	0.9385
Methionine %	0.7482	0.7162	0.7162	0.7162
Calcium %	1.5999	1.7479	1.7159	1.5979
Phosphorus %	0.4491	0.5181	0.5211	0.4211

2.2. Semen kinetics and seminal oxidative status assessment

The experimental supplementation lasted for 12 weeks, during which bucks were trained to serve artificial vagina from 9weeks. At the end of the trial, semen samples were collected from each buck, samples were centrifuged to obtain seminal plasma. Seminal fluid obtained was assayed for total protein, total antioxidant activity, lipid peroxidation, glutathione peroxidase, catalase, and superoxide dismutase activity using standard procedures [17].

The samples for semen quality assessment were evaluated for volume per buck using tuberculin syringe to the nearest 0.1 ml. After that, semen samples were evaluated for sperm kinetics using computer-assisted sperm analyser (SpermAnalyzeWin7 Xuzhou city, China, the setting of CASA as in 5th WHO manual, 51 sperm tracks, evaluated magnification x10, image acquisition rate: number frames/s 60); percentage motility, progressive motility, non-progressive motility, curvilinear velocity (um/s), average path velocity (um/s), straight-line velocity (um/s), linearity, straightness, the amplitude of lateral head (um), beat cross frequency (Hz), wobble, While sperm concentration and liveability using convention procedures. Sperm concentration (duplicates per sample) were determined using Neubauer haemocytometer (TH-100; HechtAssistant, Sondheim, Germany) and expressed as spermatozoa x10^8/ml. Liveability was done by placing a drop of semen on a glass slide, one drop of eosin–nigrosin stain added and mixed gently, then smeared on a slide, air-dried and viewed under the microscope at a magnification of x400.

2.3. Steroid hormonal assay

Fastened blood was collected from all rabbit at the end of the 12-week study, through the ear vein into a sample bottle and centrifuged to obtain serum using standard procedures. Steroid hormones were assessed using ELISA kits for follicle-stimulating hormones (FSH) using calbiotech Catalog No.: FS046F, Luteinizing hormones (LH) calbiotech ELISA Catalog No.: LH231F, Testosterone using calbiotech ELISA Catalog No. TS130S, all from Elabscience, Houston Texas, USA.

2.4. Sperm production and storage assessment

All bucks were euthanised by cervical dislocation resulting in immediate unconsciousness and death was confirmed by cessation of circulation, after which, the pairs of testes and epididymis were carefully separated and freed of all adhering connective tissues. They were separated into right and left testes, right and left Caput, corpus and caudal epididymis and weighed. The testes, Caput, corpus and caudal epididymis were homogenised separately with normal saline solution (0.154 M NaCl). The suspensions were mixed, filtered through a double layer cheesecloth and the filtrate transferred into test tubes. Sperm count was done haemocytometrically using the improved Neubauer haemocytometer.

The gonadal sperm reserve was estimated as the total number of late spermatids and spermatozoa in the homogenised testicular tissues expressed in millions.

Extra gonadal sperm reserves were determined by direct counting of sperm cells in a haemocytometer after 1:2 v/v dilutions. The extra-gonadal sperm reserve was the total number of spermatozoa in all the sections of epididymal tissues expressed in millions.

The daily sperm production was determined by direct measurement from testicular sperm reserves. The number of late spermatids and spermatozoa in homogenised testes were determined by haemocytometer counts, using time divisor of 3.43 proposed by Amann [18], as estimated from the 48-day duration of one cycle of the seminiferous epithelium**.** The daily sperm production was then calculated with the formula as outlined by Ewuola and Egbunike [19]. DSP = Testicular Sperm Count/Time Divisor.

Time divisor is 3.43 calculated by the length of one cycle of the seminiferous epithelium x Percentage of cycle represented by spermatids counted.

The efficiency of sperm production (spermatogenic efficiency) was determined by dividing the daily sperm production by the paired testes weight.

2.5. Statistical analysis

The statistical model is as below:

$$Y_{ijl} = \mu + B_i + e_{ijl}$$

Where Y_{ijl} represents the value of parameters estimated for semen quality, seminal oxidative status, sperm production and steroid hormones measured in the lth animal; μ is the overall mean for each character; B_i is the fixed effect of ith herbal supplements administered, and e_{ijl} is the random residual effect.

Data obtained from the study were subjected to the general linear model procedure of Analysis of variance using statistical analysis software [20] at $p = 0.05$. Mean differences were separated by the Duncan's Multiple Range Test (DMRT) of the same software.

3. Results

3.1. Performance of rabbits fed three herbal supplements plants

The performance of rabbits fed three herbal supplements plants is shown in Table 2. The final weight and weight gain were significantly ($p < 0.05$) lower in rabbits fed with mistletoe supplement compared to those fed with Phyllanthus supplement. Final weight and weight gain of rabbits fed control diet compared favourably with rabbits on herbal supplemented diets. However, the total dry matter intake was not significantly ($p > 0.05$) affected by supplementation with herbs. Significantly ($p < 0.05$) higher values of left epididymis weight were observed in bucks fed with Moringa supplemented diet compared to rabbits of other treatment groups. The right epididymis weight was significantly ($p < 0.05$) affected by the inclusion of herbal plants with the lowest value recorded in bucks fed the control diet. The paired epididymal weight was significantly ($p < 0.05$) higher in the group fed with the Moringa supplemented diet compared to bucks of other groups. The left testis weight was significantly ($p < 0.05$) higher in the group fed with the Moringa supplemented diet and compared favourably to the bucks fed with control diets and those of bucks fed on diets with the supplement of either mistletoe or Phyllanthus. However, the right testicular weight was significantly ($p < 0.05$) higher in the group fed with mistletoe supplemented diet and compared favourably to the bucks fed on diets with supplementation of either Moringa or Phyllanthus but better than the control group. The same trend was observed for the paired gonadal weight. The right and left caput were significantly ($p < 0.05$) higher in the bucks fed with Moringa supplemented diet compared to bucks of other groups. The left corpus weight was significantly ($p < 0.05$) higher in the group fed with either moringa or Phyllanthus supplements compared to bucks of other groups. However, the right corpus left, and right caudal of the epididymis were not significantly ($p > 0.05$) affected by the supplements.

3.2. Spermatozoa kinetics of rabbit bucks fed on diets supplemented with the three herbs

The sperm kinetics of rabbit bucks fed on diets supplemented with is three herbs is depicted in Table 3. The ejaculate volume was significantly ($p < 0.05$) affected by the supplements, with significantly ($p < 0.05$) higher values recorded in the groups fed the control diet, and Moringa supplemented diet, compared to the ejaculate of bucks in the other groups. The sperm concentration was significantly ($p < 0.05$) higher in the group fed with the mistletoe supplemented diet compared to the sperm concentration recorded in the control groups. The same trend was observed for the non-progressive motility, curvilinear velocity, average path velocity and the amplitude of lateral head. The percentage motility of spermatozoa was significantly ($p < 0.05$) higher in the group fed with mistletoe supplemented diet than those fed on the control diet and compared favourably to the bucks fed on diets with either Moringa or PhyllanthusPhyllanthus supplements. The herbal supplements linearly ($p < 0.05$) increased progressive sperm motility. However, the straight-line velocity beat cross frequency, percentage wobble liveability and acrosome integrity of spermatozoa were not significantly ($p > 0.05$) affected by the herbal supplement. The percentage spermatozoa linearity was significantly ($p < 0.05$) higher in the group fed on Phyllanthus inclusive diet and compared favourably to the bucks fed on either the control or Moringa supplemented diet but better than the buck fed on mistletoe supplemented diet. Significantly ($p < 0.05$) lower value of percentage straightness was observed in bucks fed on mistletoe supplemented diet compared to values recorded in the other groups.

3.3. Seminal oxidative status of rabbit bucks fed on herbal supplements

The seminal oxidative status of rabbit bucks fed on herbal supplements is shown in Table 4. The seminal total protein concentration was significantly ($p < 0.05$) higher in group fed with mistletoe supplement than the group fed on diet with moringa supplement. The supplementation with herbs linearly ($p < 0.05$) increased the seminal total antioxidant activity in the order of mistletoe < moringa < phyllanthus with a corresponding decrease in the seminal lipid peroxidation across the herbal supplemented treatments compared to the control. The seminal catalase

Table 2
Performance and reproductive organ weights of rabbit bucks fed herbal supplemented diets.

	Control	Moringa	Mistletoe	Phyllanthus	SEM
Initial weight (g)	776.20	777.20	779.60	773.20	58.10
Final weight (g)	1590.45^{ab}	1592.20^{ab}	1388.20^{b}	1789.45^{a}	85.37
Weight gain (g)	814.25^{ab}	815.00^{ab}	608.60^{b}	1016.25^{a}	63.29
TDM Feed intake g/d	108.30	108.20	104.25	114.0	32.46
Left Epidydermis (g)	0.48^{c}	0.81^{a}	0.49^{c}	0.61^{b}	0.07
Right Epidydermis (g)	0.49^{b}	0.64^{a}	0.57^{a}	0.57^{a}	0.05
Paired Epidydermal weight (g)	0.96^{b}	1.46^{a}	1.06^{b}	1.18^{b}	0.11
Left Testis (g)	0.49^{b}	1.17^{a}	1.06^{ab}	0.88^{ab}	0.11
Right Testis (g)	0.52^{b}	1.03^{ab}	1.20^{a}	0.79^{ab}	0.11
Left Caput (g)	0.26^{b}	0.50^{a}	0.22^{b}	0.29^{b}	0.06
Right Caput (g)	0.27^{b}	0.42^{a}	0.32^{b}	0.28^{b}	0.04
Left Corpus (g)	0.04^{b}	0.10^{a}	0.05^{b}	0.09^{a}	0.01
Right Corpus (g)	0.04	0.05	0.04	0.07	0.01
Left Caudal (g)	0.18	0.22	0.22	0.24	0.21
Right Caudal (g)	0.18	0.17	0.22	0.22	0.02
Paired gonadal weight (g)	1.00^{b}	2.20^{ab}	2.26^{a}	1.67^{ab}	0.21

ab: means along the same row with different superscripts, differ significantly (P < 0.05), SEM: standard error of mean.

concentration was significantly ($p < 0.05$) higher in the group fed on Phyllanthus supplemented diets than the group fed on the control diet and compared favourably to the bucks fed on mistletoe and Moringa supplemented diets. Seminal glutathione peroxidase activity was significantly ($p < 0.05$) lower in bucks fed on Phyllanthus supplemented diets compared to the bucks of other groups. However, seminal superoxide dismutase activity was significantly ($p < 0.05$) higher in the group fed either on Moringa or Phyllanthus supplemented diets compared to bucks of other treatments.

3.4. Steroid hormone of rabbit bucks fed on herbal supplemented diet

The steroid hormone of rabbit bucks fed on herbal supplemented diet is depicted in Table 5. Bucks fed on Moringa supplemented diet had significantly ($p < 0.05$) higher luteinising hormone level and significantly ($p < 0.05$) least testosterone level compared to bucks of other treatments. However, the follicle-stimulating hormone level was significantly ($p < 0.05$) higher in the group fed on Phyllanthus supplemented diet compared to bucks of other treatments.

3.5. Extra-gonadal sperm reserves of rabbit bucks fed on herbal supplements

The extra-gonadal sperm reserves of rabbit bucks fed on herbal supplements are shown in Table 6. The left caput sperm reserve of bucks was significantly ($p < 0.05$) higher in the group fed on mistletoe supplemented diets compared to bucks of other treatments. The left corpus sperm reserve of bucks were significantly ($p < 0.05$) lower in bucks fed on either Moringa or mistletoe supplemented diets compared to the bucks of other groups. The left caudal sperm reserve of bucks was significantly ($p < 0.05$) higher in the group fed on the control diet and decreased significantly ($p < 0.05$) across the groups. The right caput sperm reserve of bucks was significantly ($p < 0.05$) higher in the group fed on Phyllanthus supplemented diet compared to bucks of other treatments. The right corpus sperm reserve of bucks was significantly ($p < 0.05$) higher in the group fed on either Moringa or Phyllanthus supplemented diets than the groups fed onmistletoe supplemented diet and were comparable to bucks fed the control diet. The right caudal sperm reserve of bucks ranged $17.14–30.93 \times 10^7$ with significantly ($p < 0.05$) higher values recorded in the group fed on control and those fed mistletoe supplemented diets compared to bucks of moringa group. However, the total Caput and right epididymis sperm reserve of bucks were not significantly ($p > 0.05$) affected by the herbal supplements. The total corpus sperm reserve of bucks was significantly ($p < 0.05$) lower in the group fed on mistletoe supplemented diets compared to those fed on Phyllanthus supplemented diet and control groups. The total caudal sperm reserve and the left epididymis sperm reserves of bucks were significantly ($p < 0.05$) lower in herbal supplemented groups compared to the

Table 3
Semen Quality and spermatozoa kinetics of rabbit bucks fed three herbal supplemented diet.

Parameters	Control	Moringa	Mistletoe	Phyllanthus	SEM
Volume (ml)	0.42^{a}	0.41^{a}	0.32^{b}	0.32^{b}	0.02
Sperm concentration (x10^{7})	6.99^{b}	7.06^{b}	12.73^{a}	5.82^{b}	1.2
Percentage motility (%)	36.06^{b}	59.57^{ab}	71.50^{a}	62.80^{ab}	4.88
Non- Progressive motility (%)	3.03^{b}	1.51^{b}	9.00^{a}	2.82^{b}	1.09
Progressive motility (%)	32.03^{b}	58.05^{a}	61.50^{a}	62.80^{a}	5.06
Curvilinear velocity (VCL) (um/s)	3.54^{b}	4.40^{b}	8.02^{a}	3.14^{b}	0.49
Average path velocity (um/s)	3.24^{b}	3.89^{b}	7.19^{a}	3.07^{b}	0.61
Straight line velocity (um/s)	2.33	2.66	3.47	2.68	0.19
Linearity (%)	71.33^{ab}	62.65^{ab}	51.41^{b}	89.59^{a}	5.21
Straightness (%)	71.78^{ab}	68.78^{ab}	53.05^{b}	90.50^{a}	4.46
Amplitude of lateral head (um)	0.22^{b}	0.19^{b}	0.40^{a}	0.14^{b}	0.03
Beat cross frequency (Hz)	0.31	0.48	0.77	0.32	0.06
Wobble (%)	89.45	89.11	92.55	98.90	2.10
Liveability (%)	77.68	88.78	93.34	95.04	4.62
Acrosome integrity (%)	85.03	91.35	93.56	95.34	5.24

ab: means along the same row with different superscripts, differ significantly (P < 0.05), SEM: standard error of mean.

Table 4
Seminal oxidative status of rabbit bucks fed herbal supplemented diets.

	Control	Moringa	Mistletoe	Phyllanthus	SEM
Total Protein (g/dL)	11.93^{ab}	9.82^{b}	18.68^{a}	14.56^{ab}	1.28
Total antioxidant activity (mmol/litre)	0.38^{b}	0.53^{a}	0.47^{a}	0.49^{a}	0.06
Lipid Peroxidation (TBARS/mg protein)	2.48^{a}	1.38^{b}	1.73^{b}	1.13^{b}	0.18
Catalase (nm H_2O_2/min/mg protein)	3.62^{b}	5.73^{ab}	6.40^{ab}	8.18^{a}	0.75
Glutathione peroxidase (μg GSH/min/mg protein)	28.13^{a}	26.99^{a}	31.68^{a}	16.25^{b}	4.05
Superoxide Dismutase (U/min/mg protein)	1.19^{b}	1.70^{a}	1.25^{b}	1.63^{a}	0.14

ab: means along the same row with different superscripts, differ significantly ($P < 0.05$), SEM: standard error of mean.

bucks fed on the control diet.

3.6. Sperm production and gonadal sperm reserves of rabbit bucks fed on herbal supplements

The sperm production and gonadal sperm reserves of rabbit bucks fed on herbal supplements are shown in Table 7. The left testis sperm reserve per testis ranged 6.54–21.32 × 10^7 spz/testis with significantly ($p < 0.05$) higher value obtained in the bucks fed on the control diet. The left testis sperm reserve per gram testis was significantly ($p < 0.05$) least in groups fed on either Moringa or mistletoe supplemented diets. A similar trend was observed for the left testicular sperm reserve per gram testis. The right testis sperm reserve per testis was significantly ($p < 0.05$) higher in the group fed on control and mistletoe supplemented diet compared to bucks of other groups. The sperm production efficiency and right testis sperm reserve per gram testis of bucks was significantly ($p < 0.05$) least in bucks fed on Moringa supplemented diet and the significantly ($p < 0.05$) highest values were obtained in bucks on control. The testicular sperm reserve per testis was significantly ($p < 0.05$) least in bucks fed on Moringa supplemented diet and the significantly ($p < 0.05$) highest values were recorded in bucks on the control diet. The epididymal sperm reserve was significantly ($p < 0.05$) higher in the group fed control compared to the bucks fed on herbal supplemented diet. However, the daily sperm production was significantly ($p < 0.05$) least (3.96 × 10^7) in bucks fed on Moringa supplemented diet, with the highest values recorded in bucks on Phyllanthus supplemented diet (10.54 × 10^7) and control groups (12.24 × 10^7).

4. Discussion

A large number of plants have been tested for potential fertility regulatory properties [21–23]. This study compares the potency of the three herbal supplements on rabbit bucks. The growth indices of rabbits fed on medicinal plants shows that bucks fed on *Phyllanthus amarus* supplement had better growth rate among the treated bucks, while bucks fed on moringa supplement had higher reproductive organ weights across the treatments. These indicates that the enhancing potentials of the herbal supplement in bucks are results of a different mechanism. There is the previous report on the growth-promoting effect of Phyllanthus in rabbits as reported by Jimoh et al., [4]. The result of reproductive organ weight enhancement by Moringa is supported by Venkatesh et al. [24], who further affirm that the oral administration of Hydro-alcoholic extract of *Moringa oleifera* increased sexual organ weight of rats. The male reproductive organs are physiologically dependent upon testosterone via stimulating growth and secretory activity of the reproductive organs leading to increase in the number and function of somatic and germinal cells of testis and results in the increase of the testis and epididymis weight [25].

Spermatozoa kinetics of rabbit bucks fed on the herbal supplements reveal that bucks on mistletoe had better motility traits (Percentage motility, progressive motility), though was comparable to those of bucks on moringa and Phyllanthus supplements but were overall better than those of bucks with no herbal supplement. The motility, speed and trajectory traits of spermatozoa obtained from semen of bucks on Moringa and Phyllanthus supplements were similar to those not fed on supplements. Herbal supplements did not influence the viability of spermatozoa. Though, acrosome integrity and liveability of spermatozoa obtained from bucks on herbal supplements were apparently higher than bucks on control. This could suggest that beyond 12 weeks of supplementation, the effect of the herbs could be significant on spermatozoa viability. Similar to the result on *Phyllanthus amarus* fed rabbits of high spermatozoa motility and testosterone agrees with the report of Azubuike et al., [26] that male rats had higher sperm concentration, motility and increased peripheral testosterone levels following *Phyllanthus amarus* extracts gavage. Similarly, Khalifa et al. [27] reported that the presence of flavonoids in herbs had been implicated to have a role in altering androgen levels, which may also be responsible for the enhancement of male sexual behaviour. Spermatozoa concentration of semen obtained from bucks on mistletoe supplemented diet was superior to those obtained from Moringa and Phyllanthus supplemented groups. This indicates that the mistletoe supplement possesses the ingredients to enhance spermatozoa concentration and motility traits in rabbit semen. The spermatozoa speed traits (curvilinear velocity and average path velocity) were superior in mistletoe supplement fed groups. However, among the spermatozoa trajectory traits, the only amplitude of lateral head was superior in mistletoe supplemented groups. In comparison, spermatozoa trajectory trait such as linearity and straightness of spermatozoa was superior in Phyllanthus supplement fed groups.

The seminal plasma of mammals contains a complex mixture of secretions from the epididymis and various accessory sex glands [28]. The seminal protein of bucks on mistletoe supplement was superior, though was comparable with Phyllanthus supplemented

Table 5
Steroid hormones of bucks fed herbal supplemented diets.

	Control	Moringa	Mistletoe	Phyllanthus	SEM
Lutenizing Hormone (mIU/mL)	0.47^{b}	0.56^{a}	0.45^{b}	0.44^{b}	0.04
Follicle stimulating Hormone (mIU/mL)	0.47^{c}	0.52^{b}	0.49^{c}	0.67^{a}	0.07
Testosterone (ng/mL)	3.98^{a}	1.28^{b}	4.56^{a}	3.73^{a}	1.37

ab: means along the same row with different superscripts, differ significantly ($P < 0.05$), SEM: standard error of mean.

Table 6
Extra-gonadal sperm reserves of rabbit bucks fed herbal supplemented diet.

	Control	Moringa	Mistletoe	Phyllanthus	SEM
Left caput sperm reserve (x10^7 spermatozoa)	2.50^b	2.13^b	3.45^a	2.04^b	0.34
Left corpus sperm reserve (x10^7 spermatozoa)	20.11^a	11.72^b	12.92^b	16.61^a	7.38
Left caudal sperm reserve (x10^7 spermatozoa)	39.44^a	27.14^b	22.52^b	13.02^c	3.16
Right Caput sperm reserve (x10^7 spermatozoa)	1.60^b	2.01^b	1.64^b	3.96^a	0.50
Right Corpus sperm reserve (x10^7 spermatozoa)	9.73^{ab}	13.31^a	6.74^b	11.84^a	1.65
Right Caudal sperm reserve (x10^7 spermatozoa)	30.93^a	17.14^b	25.93^a	19.63^{ab}	5.64
Total caput sperm reserve (x10^7 spermatozoa)	4.10	4.14	5.10	6.00	0.56
Total corpus sperm reserve (x10^7 spermatozoa)	29.81^a	24.72^{ab}	19.60^b	28.54^a	3.84
Total caudal sperm reserve (x10^7 spermatozoa)	70.22^a	44.22^b	48.53^b	32.63^b	10.71
Right Epidydermal sperm reserve (x10^7 spermatozoa)	42.20	32.24	34.33	35.43	5.80
Left Epidydermal sperm reserve (x10^7 spermatozoa)	62.00^a	40.93^b	38.82^b	31.72^b	8.86

abc: means along the same row with different superscripts, differ significantly ($P < 0.05$), SEM: standard error of mean.

group. The inclusion of herbal supplements in bucks diets enhanced seminal total antioxidant activity and reduced lipid peroxidation. This could be attributed to catalase, and superoxide dismutase activity that was superior in Phyllanthus and Moringa supplemented bucks than bucks with no herbal supplement. As reported that antioxidant enzymes serve as a marker of oxidative imbalance, such that an increase of GPx activity in seminal plasma would be a response to the increase of ROS, in particular, hydrogen peroxide [29]. Bucks fed on mistletoe supplements had higher catalase and glutathione peroxidase activity than groups not fed on supplement. Similarly, Turk et al. [30] observed an increased spermatozoa GPx activity in rats ingesting a high dose of pomegranate juice, while low, middle and high doses of pomegranate juice increased catalase activity in rat spermatozoa.

Similarly to the mistletoe supplemented groups, El-Desoky et al. [31] reported that enhanced antioxidants and fatty acids profile of sperm cell membrane, which is essential to provide the sperm plasma membrane with fluidity, ion exchange and motility could account for increased sperm concentration, sperm motility, sperm viability, sperm morphology and sperm cell membrane integrity. Male germ cells are more susceptible to oxidative stress than somatic cells [32] because their membranes contain more polyunsaturated fatty acids. Thus, oxidative stress plays a vital role in the aetiology of sperm malformation, altered function, the sperm count profile and male infertility [33]. Similarly, Suresh et al. [34], reported a decline in seminal GPx activity in ageing rats was reversed by *M. pruriens* treatment due to normalisation in the level of glutathione and reduced ROS in the presence of the herbs. The decrease in the activity of seminal GPx as obtained in Phyllanthus supplemented bucks, maybe due to the lower level of glutathione or decreased synthesis/elevated degradation or inactivation of the enzyme in rat semen [34].

The trend of result revealed that serum testosterone peaked in bucks fed mistletoe supplemented diet. Follicle-stimulating Hormone and luteinising Hormone peaked in Phyllanthus and Moringa supplemented groups respectively. Increase in serum testosterone levels acts both centrally and peripherally in the improvement of sexual behaviour and erection [30]. This could be due to the bioactive component such as flavonoids and lignans, that may stimulate the testis or epididymis) or through a hypothalamus-pituitary-testis-axis (i.e., stimulating testosterone secretion) [30,35]. The presence of alkaloids, flavonoids, tannins, saponins, and other phytochemical constituents are well known for their ability to increase testosterone concentration [36].

Enhanced accessory sex organs and positive influence on the gonadotropins releasing hormones (GnRH) will stimulate the anterior pituitary to produce the gonadotropins hormones tend to lead to the release of LH, FSH and testosterone into the bloodstream [37]. High testosterone levels in Phyllanthus supplemented bucks agrees with Obianime and Uche [35], which affirmed that *Phyllanthus amarus* extract caused an increase in the level of testosterone in Guinea pigs. Higher FSH and LH in Moringa supplemented groups compared to control is similar to report of Khalifa et al. [27], that *M. oleifera* up-regulated the expression of fertility genes (CYP19, LH and FSH). LH reduction and increase testosterone in mistletoe and Phyllanthus supplemented bucks may be due to the negative feedback on LH synthesis because as testosterone levels rise, they have direct negative feedback on the hypothalamus and anterior pituitary gland [38]. In the arcuate and preoptic nuclei of the hypothalamus, testosterone will decrease the release of gonadotropin-releasing hormone (GnRH) hence less GnRH is released into the hypothalamic-pituitary portal system. Because of this, the anterior pituitary will release less luteinising Hormone [39].

Similarly, the herbal supplement could act directly on the anterior pituitary to inhibit the synthesis of gonadotropins, as opined by Asuquo et al. [40] that the reason for the reduction in LH and testosterone levels could be as a result of the regressing effect of the herbs on the anterior pituitary cells. Again, the result of bucks fed on moringa supplements, and the reduction in testosterone

Table 7
Sperm production and gonadal sperm reserves of rabbit bucks fed herbal supplemented diets.

	Control	Moringa	Mistletoe	Phyllanthus	SEM
Left Testis sperm reserve (x10^7 spermatozoa/testis)	21.32^a	6.54^c	11.73^{bc}	16.71^{ab}	1.89
Left Testis sperm reserve (x10^7 spermatozoa/g testis)	61.20^a	5.58^c	10.61^c	33.81^b	9.80
Right testis sperm reserve (x10^7 spermatozoa/testis)	27.43^a	7.04^b	24.40^a	9.28^b	4.66
Right testis sperm reserve (x10^7 spermatozoa/g testis)	57.84^a	6.88^c	26.03^b	18.22^b	8.11
Testisicular sperm reserve (x10^7 spermatozoa/testis	41.93^a	13.26^b	36.31^a	25.94^b	4.79
Testicular sperm reserve (x10^7 spermatozoa/g testis)	105.43^a	12.54^c	36.61^{bc}	52.03^b	13.92
Epidydermal sperm reserve (x10^7 spermatozoa)	104.00^a	73.14^b	73.22^b	67.12^b	11.63
Daily Sperm Production (spermatozoa x10^7)	12.24^a	3.96^c	10.54^a	7.56^b	1.40
Sperm Production Efficiency (spermatozoa x10^7)	30.53^a	3.63^c	10.74^b	15.24^b	4.04

abc: means along the same row with different superscripts, differ significantly ($P < 0.05$), SEM: standard error of mean.

concentration in bucks fed on *Moringa oliefera* was reported by El-Desoky et al. [31] in bucks administered with *Moringa oliefera* extract, and this suggestion may be due to the presence of diethyl phythalate, which in Moringa has shown to have an antiandrogenic activity [41]. The decreased plasma testosterone may be due to Leydig cell impairment caused by ROS generation, the inhibition of testosterone biosynthesis by Leydig cells, the inhibition of testosterone conversion to dihydrotestosterone (DHT) in DHT-sensitive tissues, the increased elimination of androgen or due to inhibition of testicular steroidogenic enzymes activity, because these enzymes are the key enzymes responsible for the regulation of testosterone biosynthesis [42].

The trend of result shows that the extra-gonadal sperm reserves of bucks fed on the herbal supplements were depleted compared to bucks, not fed on herbal supplements. Among the herbal supplement, Phyllanthus supplemented groups tend to compare favourably with the non-supplemented groups. Sexual maturation in male rabbits is related to an increase in FSH secretion since FSH binds within the seminiferous tubules to facilitate spermatogenesis [43]. Thus, a combination of increased testosterone and FSH observed in Phyllanthus supplement fed rabbits groups in this study favours spermatogenesis, which is in agreement with Alabi et al. [36], in *N. latifolia.* Increase in epididymal weight observed in bucks fed on moringa supplement with low sperm reserves could be due to parenchyma tissues resulting from an increase in ROS healthy parenchyma tissues and detrimental to sperm maturation [44].

The gonadal sperm reserves and sperm production potentials of bucks on herbal supplements decreased drastically in comparison to bucks on control. The decrease in testicular and epididymal sperm reserve observed in herbal supplemented bucks might reflect the spermatogenic cell death [42]. Bucks on mistletoe supplementation rival the superior daily sperm production and testicular sperm reserve in bucks without herbal supplements. This shows mistletoe supplemented bucks had comparable daily sperm production and testicular sperm reserve as bucks without supplement. This could be likened to Ekere et al. [2], that reported increased sperm production due to methanolic leaf extract of *D. arborea* in the albino rat. The result obtained in Moringa supplemented bucks could be due to Sertoli cells malnourishment for the growth and survival of sperm cells and inhibiting testosterone production by the Leydig cells leading to lower daily sperm production and gonadal sperm reserves [44]. Phytochemical screening of medicinal plants reveal that they possess alkaloids, saponins, tannins, terpenes, alkaloids, flavonoids, carbohydrates and cardiac glycosides, which have been implicated in male reproductive toxicology [45].

The superior daily sperm production, testicular sperm reserve, testosterone, seminal total protein and spermatozoa kinetics is revealed as a potential of mistletoe supplement over other herbs supplements. This could be due to antioxidants present in the herbal supplement, acting in concert with the antioxidant system present in the epididymis, which further preserved and enhanced the process of spermatogenesis to boost the semen quality as reported by Khalifa et al., [27]. Similar trends of different mechanism of action in spermatozoa quality from different points in the reproductive system in this study are reported by Hamilton et al. [29], which observed that differences in sperm cells quality from semen ejaculate, was absent in epididymal sperm cells following testicular heat insulation. The trend of the result of superior daily sperm production, testicular sperm reserve, seminal total protein and spermatozoa kinetics obtained in bucks fed on mistletoe supplements is explained by Ewuola et al. [44], which affirms that the spermatozoa quality and concentration are the output of the testis and that the ability to store them effectively in the epididymis are essential for successful breeding. This is similar to reports by Castellini et al. [46] which observed that flaxseed, being one of the richest sources of phytoestrogens, act as hormone-like compounds, increase sex-hormone-binding-globulin synthesis and consequently stimulating testosterone production. Moreover, Qi et al. [47] have suggested that dietary flaxseed improves semen quality by increasing the testosterone hormone secretion, which may be related to higher StAR and P450scc mRNA and SF-1 expression.

Moringa supplements in bucks' diet enhanced serum luteinising hormone and reproductive organ weights and were the superior effects of moringa supplement over other herbal supplementation in rabbit bucks' diet. Similarly, Li et al. [48] showed that dietary linseed oil supplemented during peripuberty stimulates steroidogenesis and testis development in rams. Inferior spermatozoa quality and production potentials in Moringa supplemented groups could be due to reductions in levels of testosterone and follicle-stimulating Hormone having been responsible for suppressed potency, hormonal imbalance and sexual dysfunctions in males [37].

Sperm production in testis and maturation in the epididymis is under the control of testosterone. Thus, the least sperm production and testosterone in moringa fed bucks in the present study may be due to direct damage to the Leydig cells and Sertoli cells that are directly involved in the production of spermatozoa, testosterone and may point to damaging ROS generated [42]. Testosterone is required for the attachment of different generations of germ cells in seminiferous tubules and therefore a low level of testicular testosterone may lead to detachment of germ cells from the seminiferous epithelium and may initiate germ cell apoptosis [42]. Contrary to this, is the report by Khalifa et al. [27], that antioxidants present in moringa leaves, acting in concert with the antioxidant system present in the epididymis; further preserved and enhanced the process of spermatogenesis to boost the semen quality. The result of this study indicates low spermatozoa kinetics traits, extra-gonadal, gonadal sperm reserves and sperm production potential of bucks fed on moringa supplements. Moreover, this could be likened to reports that moringa saponin induces male reproductive toxicity, via a decrease in sperm concentration and sperm motility with histomorphological damage to the Sertoli and Leydig cells [45]. Wu et al. [6], reported that epididymal epithelium could be damaged by oxidative stress and unable to protect spermatozoa during their maturation and could account for low epididymal sperm reserves in bucks fed moringa supplements.

Phyllanthus supplement in bucks' diet enhanced weight gain, serum FSH, testosterone, and extra-gonadal sperm reserves, and these were exceptional contribution of Phyllanthus supplementation to bucks' diet over other herbal supplements. Endogenous testosterone is formed from cholesterol through multiple enzymatic pathways, in the Leydig cells of the testes and the adrenal glands [49], along with its androgenic properties. It also promotes muscular growth or anabolic effects and is partly responsible for maintaining oxidative stability [50]. This could explain the growth-promoting effect of Phyllanthus supplement among other herbal supplements. The superior testosterone activity in rabbit on Phyllanthus supplements is not associated with superior spermatozoa quality in semen ejaculated and testicular sperm reserves. This could be linked to the claims by Tóthová et al. [50], which affirms that exogenous testosterone increases oxidative stress and decreases antioxidative status in rat testes. Possibly the phytohormones in the Phyllanthus supplement exhibited exogenous testosterone-like properties.

Similarly, Ogbomade et al. [51], reports that *Phyllanthus amarus* extract caused impaired sperm parameters, high levels of follicles stimulating hormone with a reduction in luteinising hormone and testosterone in rats. This indicates primary testicular failure suggestive of an increase in follicle-stimulating hormone secretion, which may principally be due to a feedback signal from damaged seminiferous tubules. LH stimulates testosterone production from the Leydig cells of the testes, without which androgen production is not possible. In contrast, FSH stimulates testicular growth and enhances the production of androgen-binding protein by the Sertoli cells, being a component of the testicular tubule necessary for sustaining the maturing sperm cell [52].

The three herbal supplements enhanced oxidative stability of rabbit semen by enhancing total antioxidant and its enzymes activities and reducing lipid peroxidation in semen ejaculates. These results may be attributed to the presence of flavonoids in the herbal supplements that can ameliorate oxidative stress-related testicular impairments in animal tissues. As corroborated by Alabi et al. [36], treatment of Male rabbits with *N. latifolia*, reduce lipid peroxidation and enhances glutathione in the semen. Similarly, Suresh et al. [34], reported a decline in seminal catalase activity in ageing rats, but was reversed by *M. pruriens* treatment, attributed to the conversion of inactive catalase to active form or increase in bioavailability of the catalase through the removal of the excess amount $O_2\cdot$ and H_2O_2 in rat semen. Despite report which reveal that testicular spermatozoa are more resistant to oxidative stress when compared to epididymal spermatozoa that flows freely in the lumen because the developing spermatozoa are guarded by the Sertoli cells that provide nutrients and antioxidant protection [6]. The result of this study reveals the depletion of sperm production in the testis of bucks fed on herbal supplements compared to the control. The trend of high seminal lipid peroxidation in control groups could be attributed its high spermatozoa production and reserve, as confirmed by Collodel et al. [53] who observed that the production of free radicals has a correlation with the intrinsic sperm release mainly because of cytosolic and mitochondrial production.

5. Conclusion

This study reveals that the inclusion of the three herbal supplements in rabbit diets enriched semen antioxidant profile and reduced its lipid peroxidation. In practical breeding programmes, the inclusion of *Moringa oleifera* supplements in bucks' diets will lead to enhance serum LH and reproductive organ weights, while *Viscum album* supplements in bucks' diets will encourage daily sperm production, testicular sperm reserves, testosterone, and spermatozoa kinetics, and *Phyllanthus amarus* supplementation in bucks' diet will enhance weight gain, serum FSH and extra-gonadal sperm reserves. The difference in the elicited response by the bucks to the three herbal supplements implies that strategic herbal supplementation would be required to meet different reproductive desires in rabbit breeding programme.

CRediT authorship contribution statement

Olatunji Abubakar Jimoh: Conceptualization, Formal analysis, Funding acquisition, Writing – original draft, All authors read and approved the final manuscript. **Wahab Adekunle Oyeyemi:** Visualization, Writing – review & editing, Project administration, Software, All authors read and approved the final manuscript. **Hafsat Ololade Okin-Aminu:** Validation, Visualization, Writing – review & editing, All authors read and approved the final manuscript. **Bolaji Fatai Oyeyemi:** co supervised the study, Data curation, All authors read and approved the final manuscript.

Declaration of competing interest

The authors of 'Reproductive potentials, semen quality, seminal oxidative status, steroid hormones, sperm production efficiency of rabbits fed *Moringa oleifera, Phyllanthus amarus* and *Viscum album* herbal supplements' wish to publish their research outcome in your reputable journal. Its content has not been published or submitted for publication elsewhere, all authors have contributed significantly and are in agreement with the content of the manuscript. We declare that no financial support or relationships those not pose conflict of interest. No conflicting interest exist in the research outcome presented in this article.

Abbreviations

Follicle stimulating hormone FSH
Lutenising hormone LH
Reactive oxygen species ROS
gonadotropin-releasing hormone – GnRH
Glutathione peroxidase GPx
Statistical analysis software SAS
Duncan's Multiple Range Test (DMRT)
Daily sperm Production DSP
Enzyme linked immunosorbent assay ELISA
Computer assisted sperm analyser CASA
World health organisation WHO
Moringa leaf meal MoLM
Phyllanthus leaf meal PLM
Mistletoe leaf meal MiLM

Ethical approval and consent to participate

The institutional committee on the care and use of animals for experiment approved this study, and it was following the NIH guide for the care and use of laboratory animals.

Consent for publication

Not applicable.

Availability of data and material

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

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